



SRI MANAKULA VINAYAGAR
ENGINEERING COLLEGE
(An Autonomous Institution)

Puducherry

SCHOOL OF PHYSIOTHERAPY

BACHELOR OF PHYSIOTHERAPY

CURRICULUM AND SYLLABI
(R-2021)



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CURRICULUM STRUCTURE

I (a). FIRST YEAR (1150 hrs)

Semester – I

Sl. No.	Course Code	Course Name	Theory (in hours)	Practical (in hours)	Total (in hours)
1.	U21BPTT11	Psychology (General and Health)	130	-	130
2.	U21BPTT12	Sociology	130	-	130
3.	U21BPTT13	Functional English	80	20	100
4.	U21BPTT14	Computer and its applications	40	60	100
Non – Examination Courses					
5.	U21BPTT15	Physiotherapy Orientation	50	-	50
6.	U21BPTP16	Physical Education	-	40	40
				Total Hours	550

Semester – II

Sl. No.	Course Code	Course Name	Theory (in hours)	Practical (in hours)	Total (in hours)
1.	U21BPTT21	Anatomy (Systemic and Regional)	150	50	200
2.	U21BPTT22	Physiology	150	30	180
3.	U21BPTT23	Nutrition	100	-	100
4.	U21BPTT24	Environmental Studies	100	-	100
Non – Examination Course					
5.	U21BPTP25	Physical Education	-	20	20
				Total Hours	600


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I (b).SECOND YEAR (1200 hrs)

Semester – III

Sl. No.	Course Code	Course Name	Theory (in hours)	Practical (in hours)	Total (in hours)
1.	U21BPTT31	General Medicine, Pediatrics and Surgery	180	-	180
2.	U21BPTT32	Microbiology and Pathology	100	-	100
3.	U21BPTT33	Biochemistry and Pharmacology	100	-	100
4.	U21BPTB34	Soft tissue manipulation	40	60	100
Non – Examination Courses					
5.	U21BPTT35	Therapeutic Yoga	10	30	40
6.	U21BPTT36	Physiotherapy ethics	20	-	20
7.	U21BPTP37	Physical Education	-	10	10
8.	U21BPTP38	Clinical Training	-	50	50
Total Hours					600

Semester – IV

Sl. No.	Course Code	Course Name	Theory (in hours)	Practical (in hours)	Total (in hours)
1.	U21BPTB41	Exercise Therapy – I	100	95	195
2.	U21BPTB42	Electrotherapy – I	100	95	195
3.	U21BPTT43	Biomechanics and Kinesiology	100	-	100
Non – Examination Courses					
4.	U21BPTT44	Basic physics for Physiotherapy	40	-	40
5.	U21BPTP45	Physical Education	-	20	20
6.	U21BPTP46	Clinical Training	-	50	50
Total Hours					600



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I (c).THIRD YEAR (1200 hours)

Semester – V

Sl. No.	Course Code	Course Name	Theory (in hours)	Practical (in hours)	Total (in hours)
1.	U21BPTB51	Exercise Therapy – II	80	100	180
2.	U21BPTB52	Electrotherapy - II	80	100	180
3.	U21BPTT53	Community Health and Rehabilitation	60	-	60
Non – Examination Courses					
4.	U21BPTP54	Physical Education	-	20	20
5.	U21BPTP55	Clinical training	-	60	60
6.	U21BPTP56	Community Health Training/ Field work	-	100	100
Total Hours					600

Semester – VI

Sl. No.	Course Code	Course Name	Theory (in hours)	Practical (in hours)	Total (in hours)
1.	U21BPTT61	Clinical Orthopedics	100	-	100
2.	U21BPTT62	Clinical Neurology	100	-	100
3.	U21BPTT63	Clinical Cardio- Thoracic Medicine	100	-	100
Non – Examination Courses					
4.	U21BPTT64	Clinical Obstetrics and Gynecology	30	-	30
5.	U21BPTT65	Radiodiagnosis for Physiotherapist	10	20	30
6.	U21BPTP66	Physical Education	-	20	20
7.	U21BPTP67	Co-curricular activities	-	20	20
8.	U21BPTP68	Clinical Training	-	200	200
Total Hours					600



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I (d) FOURTH YEAR (1200 hrs)

Semester – VII

Sl. No.	Course Code	Course Name	Theory (in hours)	Practical (in hours)	Total (in hours)
1.	U21BPTB71	Physiotherapy in Orthopedic conditions	60	60	120
2.	U21BPTB72	Physiotherapy in Neurology conditions	60	60	120
3.	U21BPTB73	Physiotherapy in Cardio-respiratory conditions	60	60	120
Non – Examination Courses					
4.	U21BPTT74	Research Methodology and Bio-statistics	30	-	30
5.	U21BPTP75	Co-curricular activities	-	10	10
6.	U21BPTP76	Clinical Training	-	200	200
Total Hours					600

Semester – VIII

Sl. No.	Course Code	Course Name	Theory (in hours)	Practical (in hours)	Total (in hours)
1.	U21BPTT81	Rehabilitation Medicine (Community and Geriatric)	90	30	120
2.	U21BPTT82	Sports Physiotherapy	60	20	80
3.	U21BPTP83	Project work	-	200	200
Non – Examination Courses					
4.	U21BPTT84	Physiotherapy in Obstetrics and Gynecology	20	20	40
5.	U21BPTT85	Clinical reasoning and evidence based practice	20	10	30
6.	U21BPTT86	Principles of Management	20	-	20
7.	U21BPTT87	Education Technology	20	10	30
8.	U21BPTP88	Clinical Training	-	80	80
Total Hours					600

COMPULSORY ROTATORY INTERNSHIP

The candidates should undergo compulsory rotatory internship training in the following departments/specialties* of Physiotherapy for the duration prescribed against each.

1.	Orthopaedics	- 20 days
2.	Neurology & Neurosurgery	- 20 days
3.	Cardiology	- 20 days
4.	Physiotherapy & Rehabilitation	- 15 days
5.	Critical care units	- 15 days
6.	General Medicine	- 15 days
7.	General & Plastic Surgery	- 15 days
8.	Obstetrics & Gynaecology	- 15 days
9.	Paediatrics & Paediatric surgery	- 15 days
10.	Community Physiotherapy	- 15 days
11.	Elective (any one of the above, or Sports Physiotherapy)	- 15 days
Total		180 days

*The above listed departments are not limited and it can be extended to any other advanced facilities available, which has to be decided by Dean, COPT.



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Syllabi (Semesters - I and II)



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PSYCHOLOGY (GENERAL AND HEALTH)

Course Code: U21BPTT11

Instruction hours: Theory – 130 hours

Course description: The course is designed to assist the students to acquire knowledge of fundamentals of psychology and develop an insight into behaviour of self and others. Further it is aimed at helping them to practice the principles of understanding the mental status and behaviour of patients in clinical settings.

General Psychology – 80 hours

Health Psychology – 50 hours

GENERAL PSYCHOLOGY [80 hours]

Unit	Hrs (T)	Content	Teaching method
I	4	INTRODUCTION Definition Schools of Psychology Methods of Psychology Branches of Psychology	Lecture Discussion
II	6	HEREDITY AND ENVIRONMENT Twins Importance of heredity and environment Role in relation to physical characteristics Intelligence and personality Nature-nature controversy	Lecture Discussion
III	8	DEVELOPMENT AND GROWTH BEHAVIOUR Infancy, Childhood, Adolescence, Adulthood, Middle age, Old age.	Lecture Discussion
IV	6	INTELLIGENCE Definitions of Intelligence Quotient, Mental Age, List of various intelligence tests – WAIS, WISC, Bhatia's performance test, Raven's Progressive Matrices test.	Lecture Discussion
V	6	MOTIVATION Definitions of motive, drive, incentive and reinforcement, Basic information about primary needs: Hunger, Thirst, Sleep, Elimination activity, Air, Avoidance of pain, Attitude to sex. Psychological needs	Lecture Discussion
VI	6	EMOTIONS Definition, differentiate from feelings, physiological changes of emotion, role of RAS, hypothalamus, cerebral cortex, sympathetic nervous system, adrenal gland, heredity and emotion, Nature and control of anger, fear and anxiety.	Lecture Discussion

Unit	Hrs (T)	Content	Teaching method
VII	10	PERSONALITY Definition List of components: Physical characteristics, character, abilities, temperament interest, attitudes. Role of heredity, nervous system, physical characteristics, abilities, family and culture on personality development. Basic concepts of Freud: Unconscious, conscious, Id, Ego and Superego. List and the define 8 stages as proposed by Erickson Concepts of learning as proposed by Dollard and Miller; drive, cue, response and reinforcement. Personality assessment Projective tests	Lecture Discussion
VIII	6	LEARNING Definition Types of learning Effective ways to learn Role of language in learning	Lecture Discussion
IX	4	THINKING Definition & creativity Creativity: Steps, traits Delusions	Lecture Discussion
X	4	FRUSTRATION Definition, sources and solution. Conflicts	Lecture Discussion
XI	8	SENSATION, ATTENTION & PERCEPTION Senses: various senses and their functions Attention: Definition, factors determining attention Perception: Definition, principles. Illusion & hallucination: types	Lecture Discussion
XII	4	LEADERSHIP Qualities and types of leadership Attitude and its changes	Lecture Discussion
XIII	4	DEFENCE MACHANISMS Defence Mechanisms of the ego List of various defence mechanisms	Lecture Discussion
XIV	4	COMMUNITY PSYCHOLOGY Social psychology Community Psychology	Lecture Discussion

HEALTH PSYCHOLOGY [50 hours]

Unit	Hrs (T)	Content	Teaching method
I	4	PSYCHOLOGICAL REACTIONS OF A PATIENT Various Psychological reactions of a patient during admission in hospital and treatment.	Lecture Discussion
II	4	REACTIONS TO LOSS Reactions to loss, death and bereavement Stages of acceptance	Lecture Discussion
III	6	STRESS Physiological and psychological changes during stress Relations to health and sickness Relaxation methods	Lecture Discussion
IV	8	COMMUNICATIONS Types of communication Elements in communications, Barriers to good communications Developing effective communication, specific communication techniques	Lecture Discussion
V	6	COUNSELLING Definition and aims Guidance and counselling Principles in counseling Personality of counsellors	Lecture Discussion
VI	4	COMPLIANCE Nature of compliance Factors contributing to non-compliance Means to improve compliance	Lecture Discussion
VII	8	EMOTIONAL NEEDS Emotional needs and psychological factors in relation to unconscious patients, handicapped persons, bed-ridden patients, patients with chronic patients, cerebral palsy children, burns, leprosy, Parkinson's disease, incontinence and mental illness.	Lecture Discussion
VIII	10	MISCELLANEOUS Geriatric psychology Paediatric psychology Behaviour modification in patients Personality styles of patients Substance abuse	Lecture Discussion

Recommended text books:

1. Ramalingam & Bid (2009). Psychology for Physiotherapists. Jaypee Brothers, New Delhi.
2. General Psychology by S.K. Mangal
3. Atkinson (1996). Dictionary of Psychology.

SOCIOLOGY

Course Code: U21BPTT12

Instruction hours: Theory – 130 hours

Course description: The course is designed to introduce the basics of sociological concepts, principles and social process, social institutions in relation to individual, family and community in India and its relationship with health, illness and handicap.

Unit	Hrs (T)	Content	Teaching method
I	6	INTRODUCTION Definition Sociology – a science of society Application of sociology in physiotherapy	Lecture Discussion
II	14	SOCIOLOGY AND HEALTH Social factors affecting health status Social consciousness and meaning of illness Perception of illness Decision making in taking treatment Institutions of health and their role in the improvement of health of the people	Lecture Discussion
III	10	SOCIALISATION Meaning of socialisation Influence of social factors on personality Socialisation in hospitals Socialisation in rehabilitation of patients	Lecture Discussion
IV	10	SOCIAL GROUPS Concept of social group Influence of formal and informal groups on health on health and sickness Role of primary and secondary groups in the hospital and rehabilitation settings.	Lecture Discussion
V	16	FAMILY & COMMUNITY Influence of family on human personality Changes in the functions of a family Influence of the family on the individual's health, family and nutrition Effects of sickness on family, family and psychosomatic disease Concept of community Role of rural and urban communities in public health Role of community in determining beliefs, practices and home remedies in treatment	Lecture Discussion



Unit	Hrs (T)	Content	Teaching method
VI	10	CULTURE & CASTE SYSTEM Components of culture Impact of culture on human behaviour Cultural meaning & response of sickness, Choice of treatment Culture induced symptoms and disease Sub-culture of medical workers Caste system: Features of modern caste system & its trends	Lecture Discussion
VII	14	SOCIAL CHANGE Meaning of social change Factors of social change on human adaption, stress, deviance and health programmes Role of social planning in the improvement of health and rehabilitation.	Lecture Discussion
VIII	12	SOCIAL CONTROL Meaning of social control, Role of norms Folkways, customs, morals, religion, law and other means of social control in the regulation of human behaviour. Social deviance and disease.	Lecture discussion
IX	12	SOCIAL PROBLEMS OF THE DISABLED Consequences of the following social problems in relation to sickness and disability. Remedies to prevent the following problems: Population explosion, poverty and employment, beggary, juvenile delinquency, prostitution, alcoholism, problems of women in employment.	Lecture Discussion
X	16	SOCIAL SECURITY Social security and social legislation in relation to the disabled.	Lecture Discussion
XI	10	SOCIAL WORKER Role of a medical social worker	Lecture Discussion

Recommended Books:

1. Bid D. (2006). Sociology for Physiotherapists. Jaypee Brothers, New Delhi.
2. Sachdeva and Vidyabushan: Introduction to the study of Sociology.
3. K. Parks Textbook of Preventive & Social Medicine.
4. Textbook of Preventive & Social Medicine – P.K. Mahajan & M.C.Gupta

FUNCTIONAL ENGLISH

Course Code: U21BPTT13

Instruction hours: Theory – 80 hours
Practical – 20 hours

Course description: The course is designed to enable to enhance ability to comprehend spoken and written English (and use English) required for effective communication in their professional work. Students will practice their skills in verbal and written English during clinical and classroom experiences.

Unit	Hrs (T+P)	Content	Teaching method
I	10+3	INTRODUCTION Study techniques Logical processes of analysis and synthesis Use of dictionary Effective diction	Lecture, Demonstration & Exercises to students
II	15+3	APPLIED GRAMMAR Review of grammar & correct usage Building vocabulary Structure of sentences & paragraphs Phonetics Public speaking	Lecture, Demonstration, Conversation & Public speaking
III	15+5	FORMS OF COMPOSITION Letter writing Note taking Précis writing Essay writing Anecdotal records Diary writing Reports Resume / Curriculum vitae etc.	Demonstration & Exercises to students
IV	15+3	COMMUNICATION Oral report Discussion Lecture / seminar Debate Summary Telephonic conversation	Demonstration & Exercises to students
V	10+3	READING COMPREHENSION Selected materials, articles, magazines, journals etc.	Demonstration & Exercises to students
VI	15+3	LISTENING COMPREHENSION Media, Audio, Video, Speeches etc.	Demonstration & Exercises to students

Recommended Books:

1. Communicative English for Engineers and Professionals. Author, Nitin Bhatnagar
2. English for physiotherapy, Joanna Ciecierska

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COMPUTER AND ITS APPLICATIONS

Course Code: U21BPTT14

Instruction hours: Theory – 40 hours
Practical – 60 hours

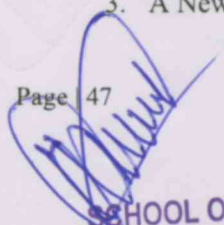
Course description: This course is designed for students to develop basic knowledge of fundamentals of computer and its application in Physiotherapy.

Unit	Hrs (T+P)	Content	Teaching method
I	4+0	INTRODUCTION TO COMPUTERS Concepts & features of computer Application areas of computers in health services Hardware and software	Lecture
II	8+15	HARDWARE Architecture of computers Types of storage devices Characteristics of disks, terminals, printers, network etc. Disk operating system: DOS, Windows Applications of networking concepts	Lecture Discussion Demonstration & Practicals
III	8+15	SOFTWARE Classification of software Application of software Operating system, computer system Computer virus: Precautions & dealing	Lecture Demonstration & Practicals
IV	10+20	PROGRAMMES MS – Word MS – Excel with pictorial presentations MS – Access MS – PowerPoint	Lecture Demonstration & Practicals
V	10+10	COMPUTER APPLICATIONS Multimedia: Types & uses Computer aided teaching & testing Use of internet: web pages & e-mail Principles in scientific research: Work processing, Health care systems, libraries, education, information system Application in Physiotherapy: E.M.G., Biofeedback, Exercise testing equipments, Spirometry, etc.	Visit, Demonstration & Practicals

Recommended Books:

1. Basic Computer Knowledge Fundamental of Computer system- Reema Thareja
2. Basic Computer Knowledge - Maluth John Monyjok
3. A New Approach to Basic Computer Education - D. P. Nagpal

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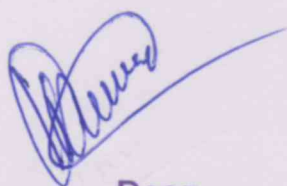
PHYSIOTHERAPY ORIENTATION

Course Code: U21BPTT15

Instruction hours: Theory – 50 hours

Course description: The course is designed to help the students to develop an understanding of the philosophy, objectives and process of physiotherapy in various clinical settings. It is aimed at helping the students to acquire knowledge, understanding and skills in physiotherapy techniques in clinical settings.

Unit	Hrs (T)	Content	Teaching method
I	5	INTRODUCTION TO HEALTH Health and Health care delivery system	Lecture
II	10	INTRODUCTION TO HEALTH SCIENCE Overview of Health Science, Health Professions & their specialties	Lecture
III	10	PHYSIOTHERAPY PROFESSION History of Medical Therapeutics History of Physiotherapy Overview of impairment, disability, handicap Health – levels of prevention & Rehabilitation Physiotherapy in medical rehabilitation	Lecture, Demonstration
IV	10	PHYSIOTHERAPY IN MEETING HEALTH CARE NEEDS OF INDIA Needs versus demands Need for physiotherapy Scope of the profession Role of Physiotherapist in health care delivery system and prevention of disability	Lecture, Demonstration, Group discussions
V	15	PHYSIOTHERAPEUTIC METHODS Physical agents in therapy Exercise therapy Electrotherapy Specialties in physiotherapy Areas of physiotherapy services & training	Lecture Demonstration & Visit



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PHYSICAL EDUCATION

Course Code: U21BPTT16

Instruction hours: – 40 hours Practical

Course description: The purpose of the course is to acquire knowledge and understand various components in physical fitness and training methods.

Unit	Hrs (P)	Content	Teaching method
I	5	INTRODUCTION Physical fitness	Lecture Discussion
II	15	TRAINING METHODS Definition Motor component Warming up Conditioning Cool down	Lecture Discussion & Demonstration.
III	20	PHYSICAL FITNESS & TRAINING Physical Fitness and Training of Motor components: Strength Speed Endurance Mobility Co-ordination	Lecture Discussion & Demonstration.



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ANATOMY (SYSTEMIC AND REGIONAL)

Course code: U21BPTT21

Instruction hours: Theory –150hours
Practical –50 hours

Course description: The course is designed to enable the students to acquire knowledge of normal structure of various human body systems particularly on musculoskeletal, nervous and cardio-pulmonary systems and understand their application in the practice of physiotherapy.

I - SYSTEMIC ANATOMY [70 HOURS]

Unit	Hrs (T+P)	Content	Teaching method
I	6+0	INTRODUCTION TO ANATOMICAL TERMS Definitions, subdivisions, systems of the body Cell: Structure, composition, function, cell division Tissues: Definition, types, characteristics, classification, location, functions. Genes and chromosomes.	Lecture Demonstration
II	8+0	CARDIO VASCULAR SYSTEM Structure of heart, blood vessels Blood and nervous supply of the heart Major blood vessels	Lecture Demonstration
III	2+0	LYMPHATIC SYSTEM Structure of Lymphatic organs and vessels Functional roles	Lecture Demonstration
IV	8+0	RESPIRATORY SYSTEM Structure of the organs of the respiratory system Muscles of respiration, tracheobronchial tree, bronchopulmonary segments	Lecture Demonstration
V	5+0	DIGESTIVE SYSTEM Structure of the alimentary tract and organs of digestive system Anatomy of the liver and pancreas	Lecture Demonstration
VI	3+0	GENITO-URINARY SYSTEM Structure of the organs of the genito-urinary system	Lecture Demonstration
VII	3+0	ENDOCRINE SYSTEM Structure of endocrine glands	Lecture Demonstration
VIII	20+0	NERVOUS SYSTEM Division of the nervous system and their organs Structure and functions of nerve cell Structure of brain, spinal cord and peripheral nerves (in detail) Structure & location of autonomic nervous system	Lecture Demonstration

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Unit	Hrs (T+P)	Content	Teaching method
IX	5+0	OSTEOLOGY Definition and types of skeletal system Classification of bones Ossification: definition, types and process	Lecture Discussion, Demonstration
X	5+0	ARTHROLOGY Definition and classification of joint Functions of joints: mobility & stability	Lecture Discussion, Demonstration
XI	5+0	MYOLOGY Structure and types of muscles Skeletal muscles: classification, forms & groups Position, origin, insertion, nerve supply and action of skeletal muscles	Lecture Discussion, Demonstration

II - REGIONAL ANATOMY [Theory-80 Hours, Practical-50 Hours]

Unit	Hrs (T+P)	Content	Teaching method
XII	25+15	UPPER EXTREMITY Osteology, arthrology, myology of the following: Pectoral region Scapular region Axilla Shoulder girdle and arm Elbow and forearm Wrist and hand Nerves of upper limb Blood vessels of upper limb	Lecture Demonstration & Practicals
XIII	25+15	LOWER EXTREMITY Osteology, arthrology, myology of the following: Pelvic & gluteal region Hip & thigh region Knee & leg Ankle & foot Nerves & Blood vessels of lower limb	Lecture Demonstration & Practicals
XIV	15+10	TRUNK Osteology, Arthrology, myology and their relations of: Vertebral Column Thoracic cage Abdomen Pelvis	Lecture Demonstration & Practicals
XV	15+10	HEAD & NECK Musculoskeletal and neurovascular features of neck and cranium Cranial nerves	Lecture Demonstration & Practicals

PRACTICALS AND DEMONSTRATION:

1. Upper extremity including surface Anatomy. Demonstration of the muscles of the upper extremity, movements in joints, identification of body prominences on inspection and by palpation, points of palpation of nerves and arteries. Identification of the bones of the upper extremity, side determination, parts, attachment of the muscles, nerves and vessels relation to bone.
2. Lower extremity including surface Anatomy. Demonstration of the muscles of the lower extremity, movements in joints, identification of body prominences on inspection and by palpation, points of palpation of nerves and arteries. Identification of the bones of the lower extremity, side determination, parts, attachment of the muscles and relation of nerves and vessels to bone.
3. Demonstration of the Head & Neck and Spinal cord & Brain including surface Anatomy.
4. Demonstration of the muscles of the back, pelvic girdle, pre and para vertebral muscles, movements in joints, identification of body prominences on inspection and by palpation.
5. Identification of the bones of the vertebral column (cervical, thoracic, lumbar, sacral and coccygeal) parts, attachment of the muscles and relation of nerves and vessels to bone.
6. Surface Markings of Various Organs and Bony Prominences
7. Radiographic Identification of Bone and Joints

Recommended Text books:

1. SNELL [Richard S], Clinical Anatomy for Medical students: Ed. 5. Little Brown and Company Boston.
2. B.D Chaurasia's Human Anatomy – Regional and Applied; Volume I, Volume II and Volume III.
3. SINGH [Inderbir], Human Osteology. JP Brothers, New Delhi 1990.
4. SINGH [Inderbir], Text book of Anatomy with colour atlas: Vol I, II, III.
5. SINGH [Inderbir], Essentials of Anatomy JP Brothers, New Delhi
6. Anatomy and Physiology - Dr. Minakshi Pathak
7. The carnial Nerves Anatomy, Pathology, Pathophysiology Diagnosis Treatment- M. Samii, P.J Jannetla
8. Anatomy by Vishram Singh

Recommended Text books for Practical:

1. ROMANES [G J], Cunningham manual of practical anatomy: Vol I, II, III

Reference Books:

1. PODAR - Handbook of Osteology: Ed. 11 Scientific book co.
2. Gray's Anatomy
3. McMinn – McMinn's color atlas of Human Anatomy.



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PHYSIOLOGY

Course Code: U21BPTT22

Instruction hours: Theory – 150 hours
Practical –30 hours

Course description: The course is designed to assist the students to acquire knowledge of normal physiology of various human body systems and understand the alterations in physiology in diseases for physiotherapy practice.

Unit	Hrs (T+P)	Content	Teaching method
I	10+0	CELL PHYSIOLOGY Cell: Structure & functions of components Functions of membranes & glands	Lecture
II	15+6	CIRCULATORY SYSTEM Blood: Component and their functions, blood groups, coagulation, blood volume and its regulation Functions and regulations of the heart, cardiac cycle, cardiac output, E.C.G., heart sounds. Blood pressure: Maintenance and regulation. Effects of exercises on postural changes.	Lecture Discussion, Demonstration & Practicals
III	20+6	RESPIRATORY SYSTEM Functions of the respiratory organs Physiology of respiration Pulmonary ventilation, volume Mechanics of respiration Gaseous exchange in lungs Regulation of respiration Effects of exercises on respiration	Lecture Discussion, Demonstration & Practicals
IV	10+0	DIGESTIVE SYSTEM Functions of organs of digestive tract Movements of the alimentary tract Digestion in mouth, stomach, intestines Absorption of food Metabolism of carbohydrates, proteins and fat	Lecture Discussion, Demonstration
V	10+0	EXCRETORY SYSTEM Functions of organs of excretory tract Composition of urine Mechanism of urine formation & Micturition Functions of skin	Lecture Discussion, Demonstration
VI	10+0	ENDOCRINE SYSTEM Functions of the various endocrine glands Endocrine Hormones: Functions and their abnormalities.	Lecture Discussion, Demonstration

Unit	Hrs (T+P)	Content	Teaching method
VII	10+0	REPRODUCTIVE SYSTEM Functions of male reproductive system Functions of female reproductive system Outline of pregnancy, parturition, lactation Contraceptive measures Physiology of foetal growth	Lecture Discussion, Demonstration
VIII	20+6	NERVOUS SYSTEM Properties and functions of Neuron Mechanism of Stimulus and nerve impulse Functions of brain, spinal cord, cranial and spinal nerves. Synaptic transmission, reflexes, control of postures and voluntary motor activity. Autonomic Nervous System	Lecture Discussion, Demonstration & Practicals
IX	5+2	SENSORY ORGANS Functions of the skin, eye, ear, nose and tongue	Lecture Discussion, Demonstration & Practicals
X	20+6	MUSCULAR SYSTEM Microscopic structure of muscle tissue, myoneural junction Physiology of Muscle contraction Exercise metabolism Muscular activity based on metabolism and fatigue Physiological changes on aging Exercise physiology	Lecture Discussion, Demonstration & Practicals
XI	20+4	APPLIED PHYSIOLOGY Heart and circulation: Normal ECG, blood pressure, cardiovascular compensation for postural and gravitational changes, determinants of cardiac performance. Neuromuscular system: Degeneration and re-generation of nerves, control of posture and voluntary movement, neuromuscular transmission, electrical phenomenon. Respiratory system: Normal breath sound, volume and compliance, effects of exercise on respiration, artificial respiration.	Lecture Discussion, Demonstration & Practicals



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Unit	Hrs (T+P)	Content	Teaching method
VII	10+0	REPRODUCTIVE SYSTEM Functions of male reproductive system Functions of female reproductive system Outline of pregnancy, parturition, lactation Contraceptive measures Physiology of foetal growth	Lecture Discussion, Demonstration
VIII	20+6	NERVOUS SYSTEM Properties and functions of Neuron Mechanism of Stimulus and nerve impulse Functions of brain, spinal cord, cranial and spinal nerves. Synaptic transmission, reflexes, control of postures and voluntary motor activity. Autonomic Nervous System	Lecture Discussion, Demonstration & Practicals
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NUTRITION

Course Code: U21BPTT23

Instruction hours: Theory – 100 hours

Course description: The course is designed to assist the students to acquire knowledge of nutrition for maintenance of optimum health and its application for different ages, activities in metabolic disorders.

Unit	Hrs (T)	Content	Teaching method
I	10	FOOD & NUTRITION Introduction Nutrition: Concepts & various aspects Role of nutrition in healthy body National nutritional policy Food: Role in nutritional & medicinal values Elements of nutrition: Macro & micro nutrients Calorie & Basal Metabolic Rate	Lecture Discussion
II	20	CARBOHYDRATES, PROTEINS, FATS Classification & caloric value Recommended daily allowance Dietary sources Functions Digestion, Absorption & Storage Malnutrition: Deficiencies & Over consumption	Lecture Discussion
III	15	WATER & ELECTROLYTES Water: Daily requirement, sources, regulation of water Metabolism Electrolytes: Types, sources, composition of body fluids	Lecture Discussion
IV	20	VITAMINS & MINERALS Classification Recommended daily allowance Dietary sources Functions Absorption and storage Deficiencies & Hypervitaminosis	Lecture Discussion
V	15	ENERGY Requirements of different categories of people Measurement of energy Body Mass Index and basic metabolism Basal Metabolic Rate – determination and factors affecting it	Lecture Discussion



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Unit	Hrs (T)	Content	Teaching method
VI	20	BALANCED DIET Concept Recommended Daily Allowance Nutritive value of foods Planning balanced diets for different categories of people Budgeting of food	Lecture Discussion

Recommended text books:

1. Comprehensive Textbook of Nutrition for BSc Nurses- Rishi Avasthi



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ENVIRONMENTAL STUDIES

Course Code: U21BPTT24

Instruction hours: Theory – 100 hours

Course description:

The course is designed as per the UGC regulation for all under graduate courses of branches of higher education. The subject is designed to refresh the student regarding the multidisciplinary nature of the environment and conservation of the ecosystem.

Course Objectives: at the end of the course, the candidate should

1. Know about the environment
2. Understand the surrounding
3. Know about the biotic interaction

Unit	Hrs (T)	Content	Teaching method
I	6	THE MULTIDISCIPLINARY NATURE OF ENVIRONMENT STUDIES: • Definition, scope and importance • Need for public awareness.	Lecture Discussion
II	18	RENEWABLE AND NON-RENEWABLE RESOURCES • Forest resources: Use and over-exploitation, deforestation, case studies, timber extraction, mining, dams, and their effects on forests and tribal people. • Water Resources: Use and over-utilization of surface and ground water, floods drought, conflicts over water, dam benefits and problems. • Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies • Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer – pesticide problems, water logging, salinity, case studies.	Lecture Discussion

Unit	Hrs (T)	Content	Teaching method
		- Energy resources: Growing energy needs, renewable and non-renewable energy resources, use of alternate energy sources, case studies. - Land resources: Land as a resource, land degradation, man induces landslides, soil erosion and desertification Role of an individual in conservation of natural resources. Equitable use of resources for sustainable lifestyles.	Lecture Discussion
III	16	ECOSYSTEMS - Concept of an ecosystem - Structure and function of an Ecosystem - Producer, consumers, and decomposers - Energy flow in the ecosystem - Ecological succession Food chains, food webs and ecological pyramids Introduction, types, characteristic features, structure and function of the ecosystem Forest ecosystem, Grassland ecosystem, Desert ecosystem, Aquatic ecosystem (ponds, streams, lakes, estuaries)	Lecture Discussion
IV	14	BIODIVERSITY AND ITS CONSERVATION Introduction – Definition of genetics, species, and ecosystem diversity Biogeographical classification of India Value of Biodiversity: consumptive use, productive use, social, ethical aesthetic, and option views.	Lecture Discussion

Unit	Hrs (T)	Content	Teaching method
V	14	ENVIRONMENTAL POLLUTION: • Air pollution • Water Pollution • Soil Pollution • Marine Pollution • Noise Pollution • Thermal Pollution • Nuclear hazards Soil waste management: Causes, effects and control measures of urban and industrial wastes. Role of an individual in prevention of pollution. Pollution case studies. Disaster management: floods, earthquake, cyclone and landslides.	Lecture Discussion
VI	16	SOCIAL ISSUES AND THE ENVIRONMENT • From Unsustainable to Sustainable development • Urban problems related to energy • Water conservation, rain water harvesting, watershed management • Resettlement and rehabilitation of people; its problems and concerns. Case Studies • Environmental ethics: Issues and possible solutions. • Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust. Case Studies. • Wasteland reclamation. • Consumerism and waste products. • Environment Protection Act. • Air (Prevention and Control of Pollution) Act • Water (Prevention and control of Pollution) Act • Wildlife Protection Act • Forest Conservation Act • Issues involved in enforcement of environmental legislation. • Public awareness	Lecture Discussion

Unit	Hrs (T)	Content	Teaching method
VII	16	HUMAN POPULATION AND THE ENVIRONMENT • Population growth, variation among nations. • Population explosion – Family Welfare Programme. • Environment and human health. • Human Rights. • Value Education. • HIV/AIDS • Women and Child Welfare. • Role of Information Technology in Environment and human health • Case Studies.	Lecture Discussion

Recommended text books:

1. Textbook of Environmental Studies for Undergraduate Courses-Erach Barucha

Reference:

1. Agarwal, K.C.2001 Environmental Biology, Nidhi Publications Ltd. Bikaner
2. Odum, EP.1971 Fundamentals of Ecology. W B Saunders Co



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PHYSICAL EDUCATION

Course Code: U21BPTT25

Instruction hours: – 20 hours Practical

Course description: The purpose of the course is to acquire knowledge and understand various components in physical fitness and training methods.

Unit	Hrs (P)	Content	Teaching method
I	2	INTRODUCTION Sports and Wellness	Lecture Discussion
II	10	TRAINING METHODS Imitating Method Visual Method PERIODIZATION 1. Preparatory phase 2. Competitive phase 3. Relaxation or Recovery phase	Lecture Discussion Demonstration & Practicals.
III	8	ACTIVITIES Weight Lifting Weight Training	Lecture Discussion Demonstration & Practicals.



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